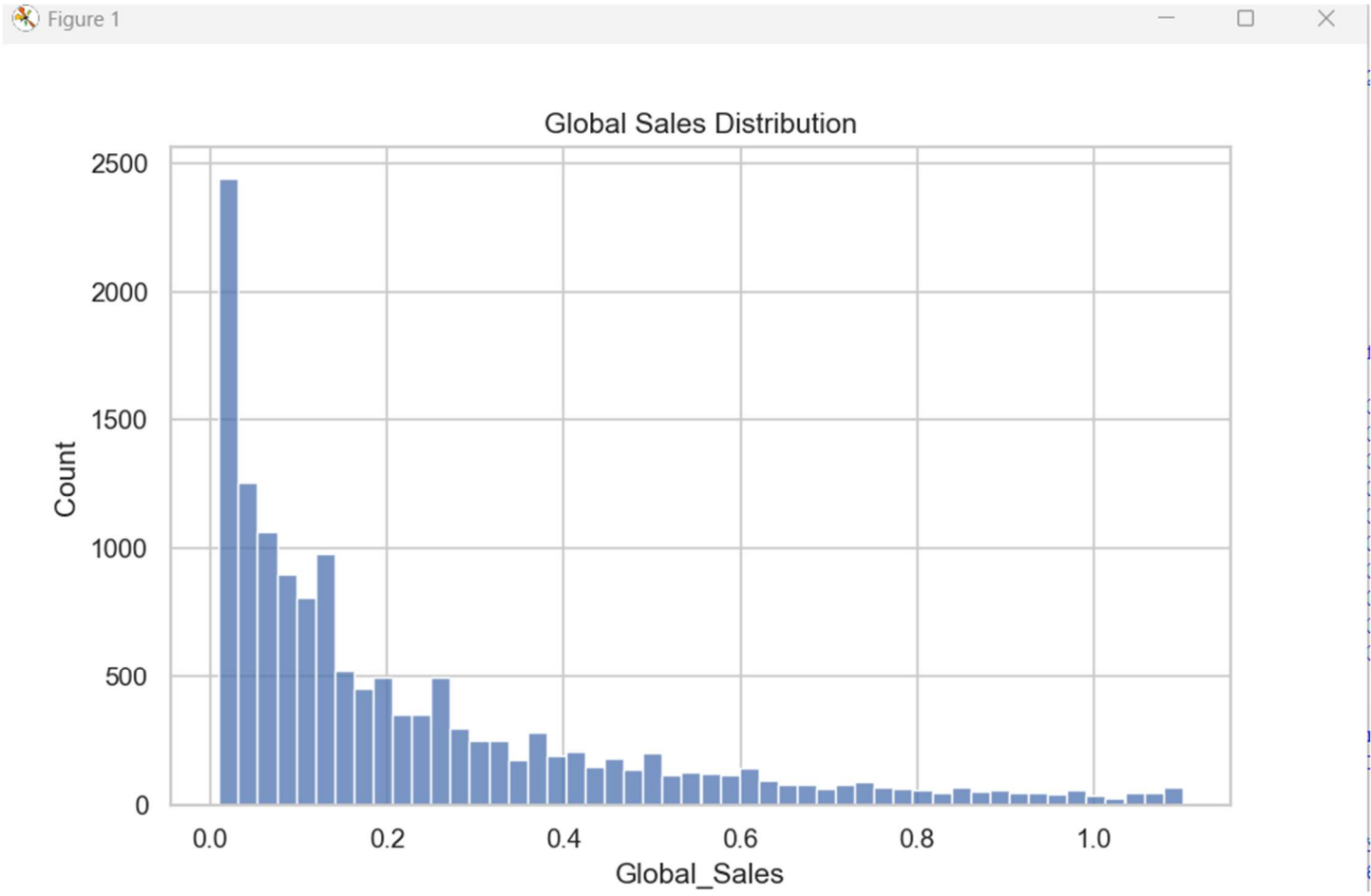
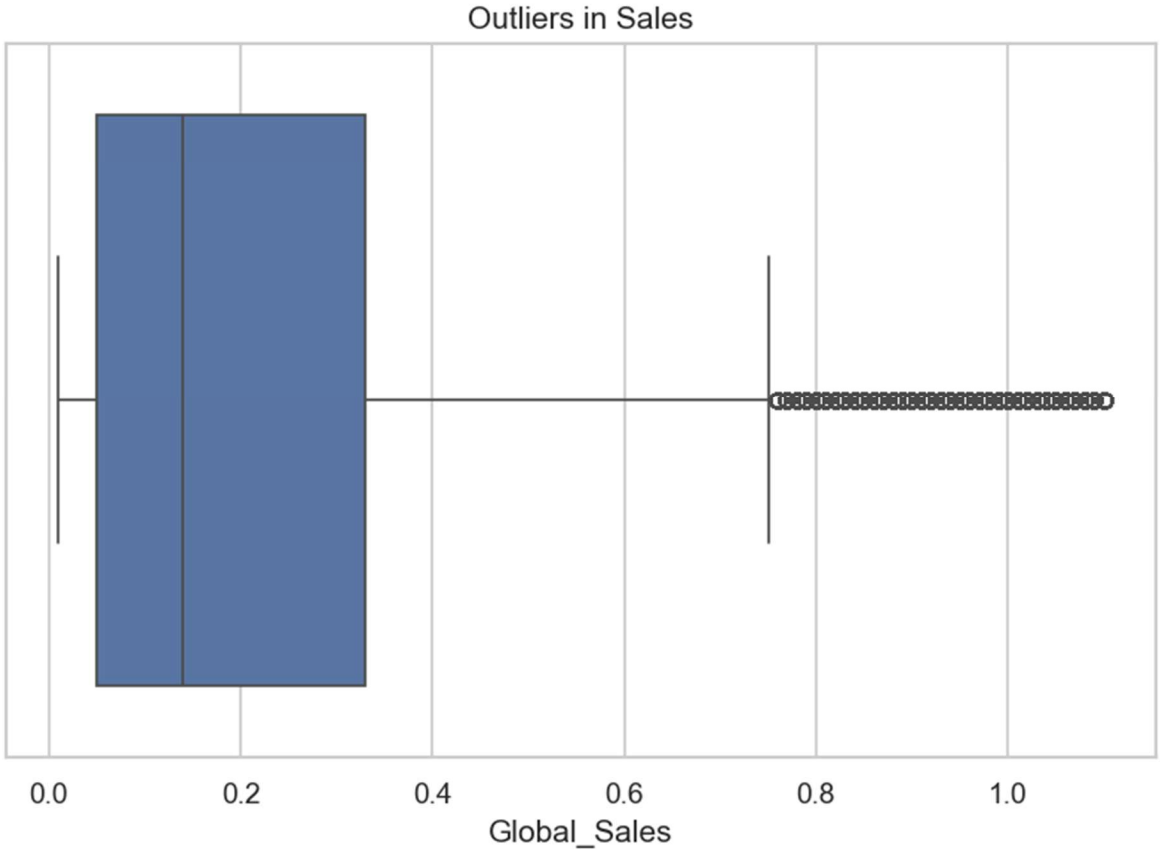
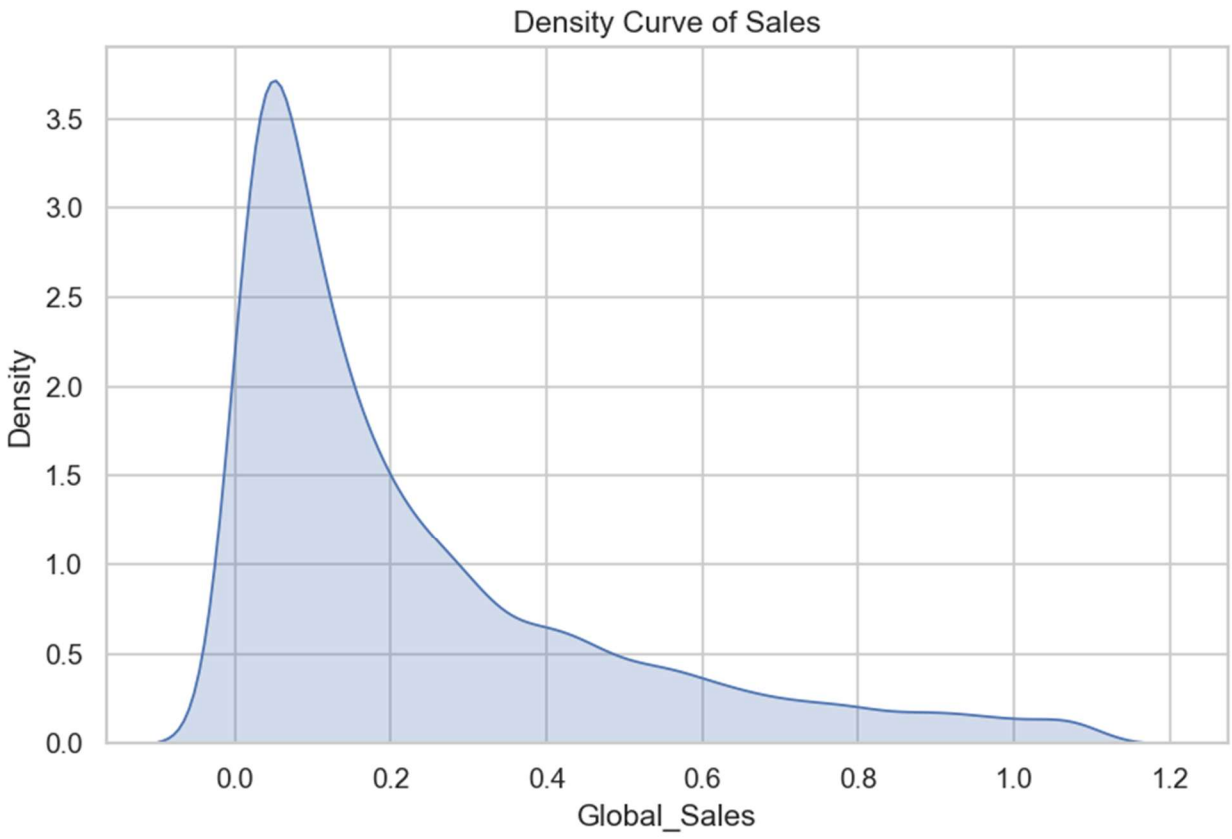
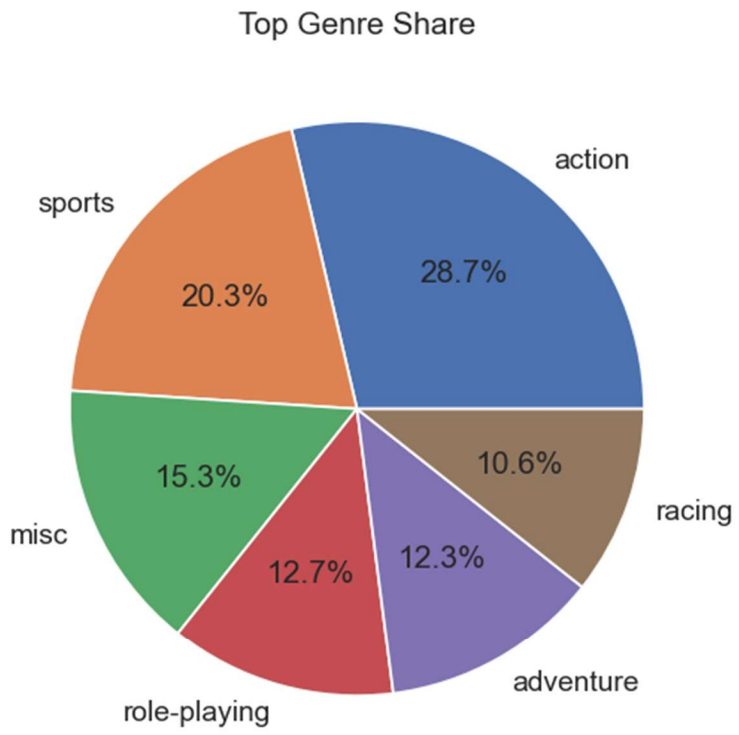
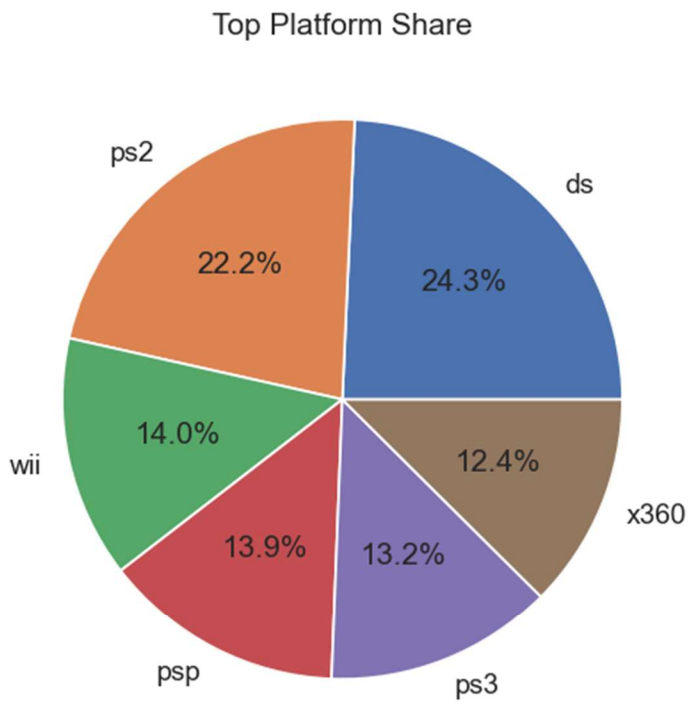


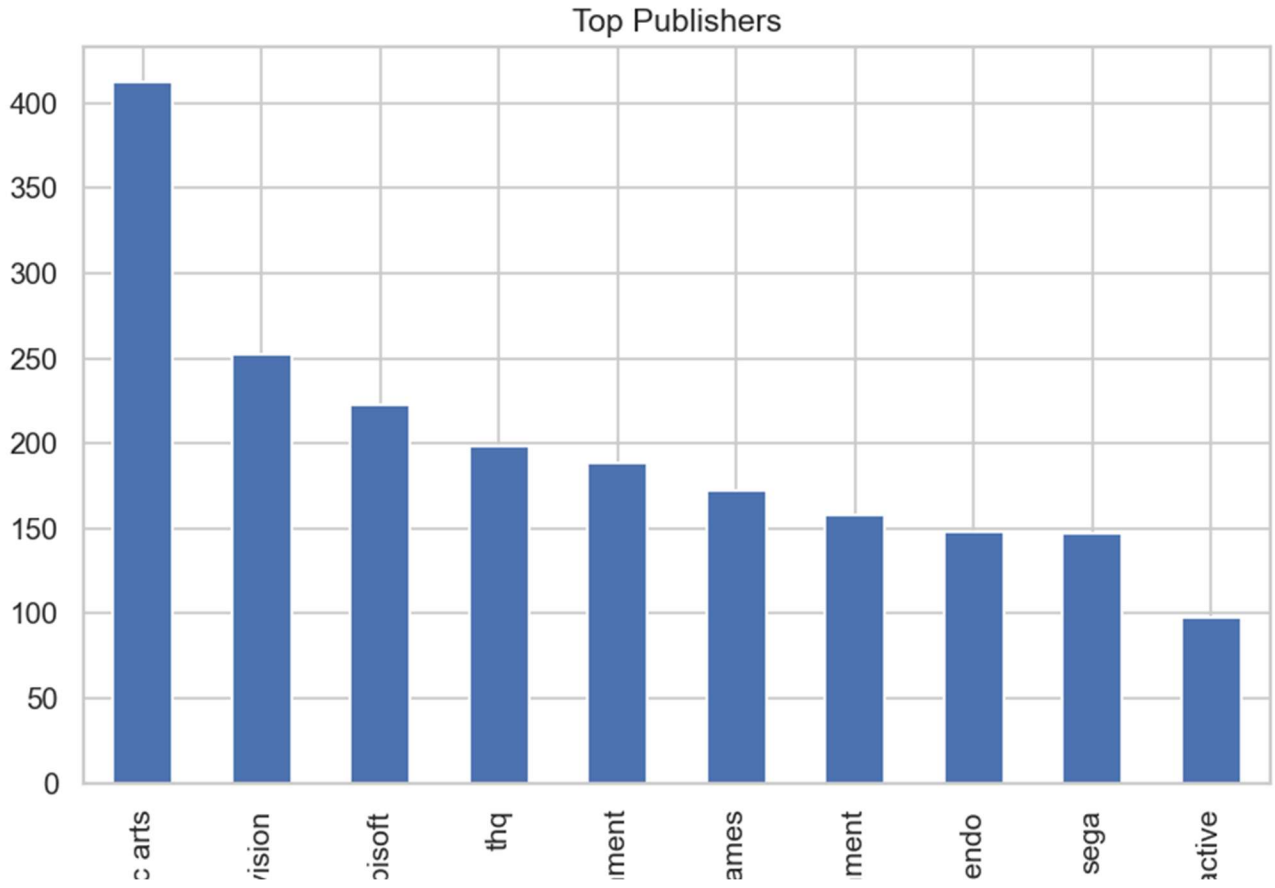
Figure 1
Global Sales Distribution
Count
2500
2000
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0
Global_Sales

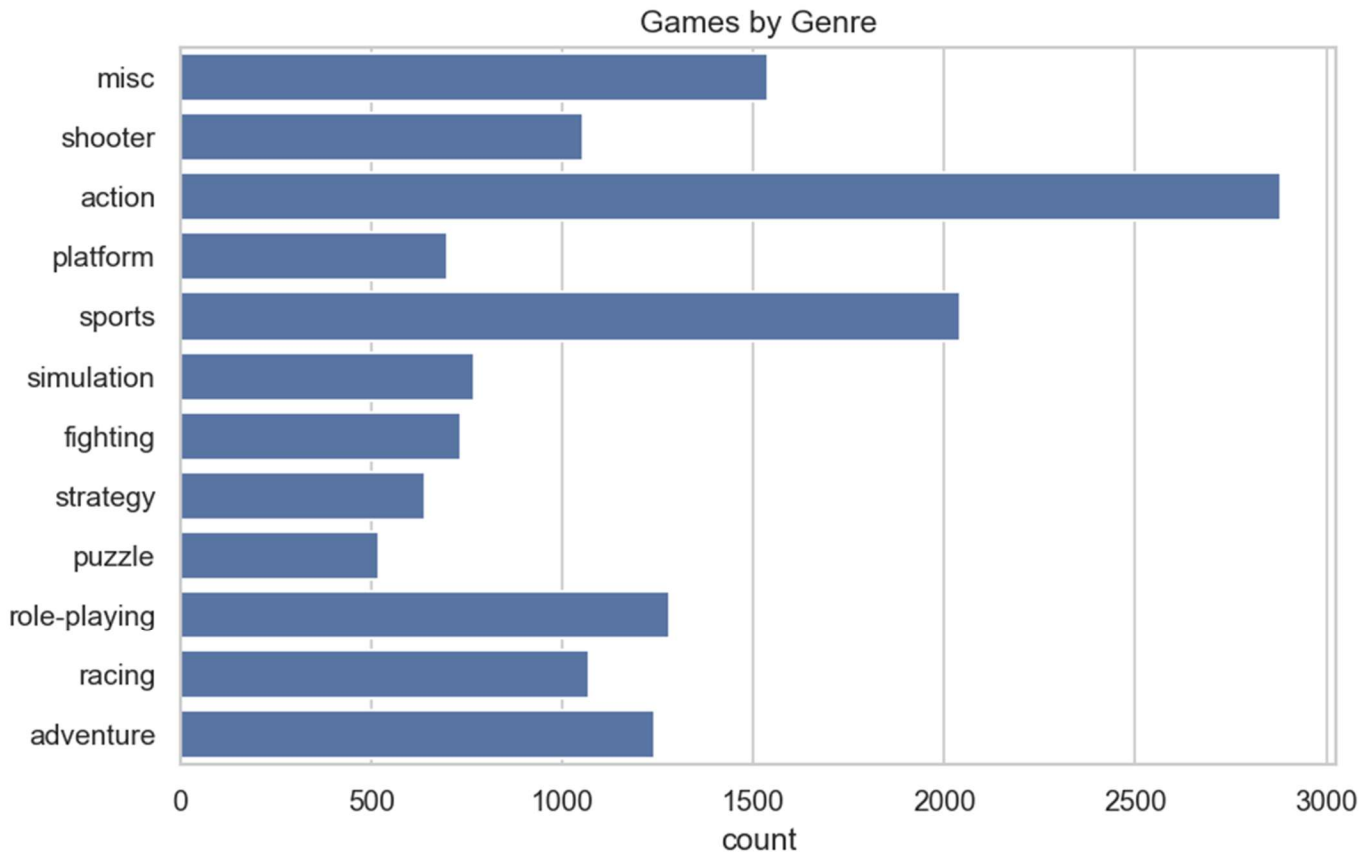

Outliers in Sales
0.0
0.2
0.4
0.6
0.8
1.0
Global_Sales

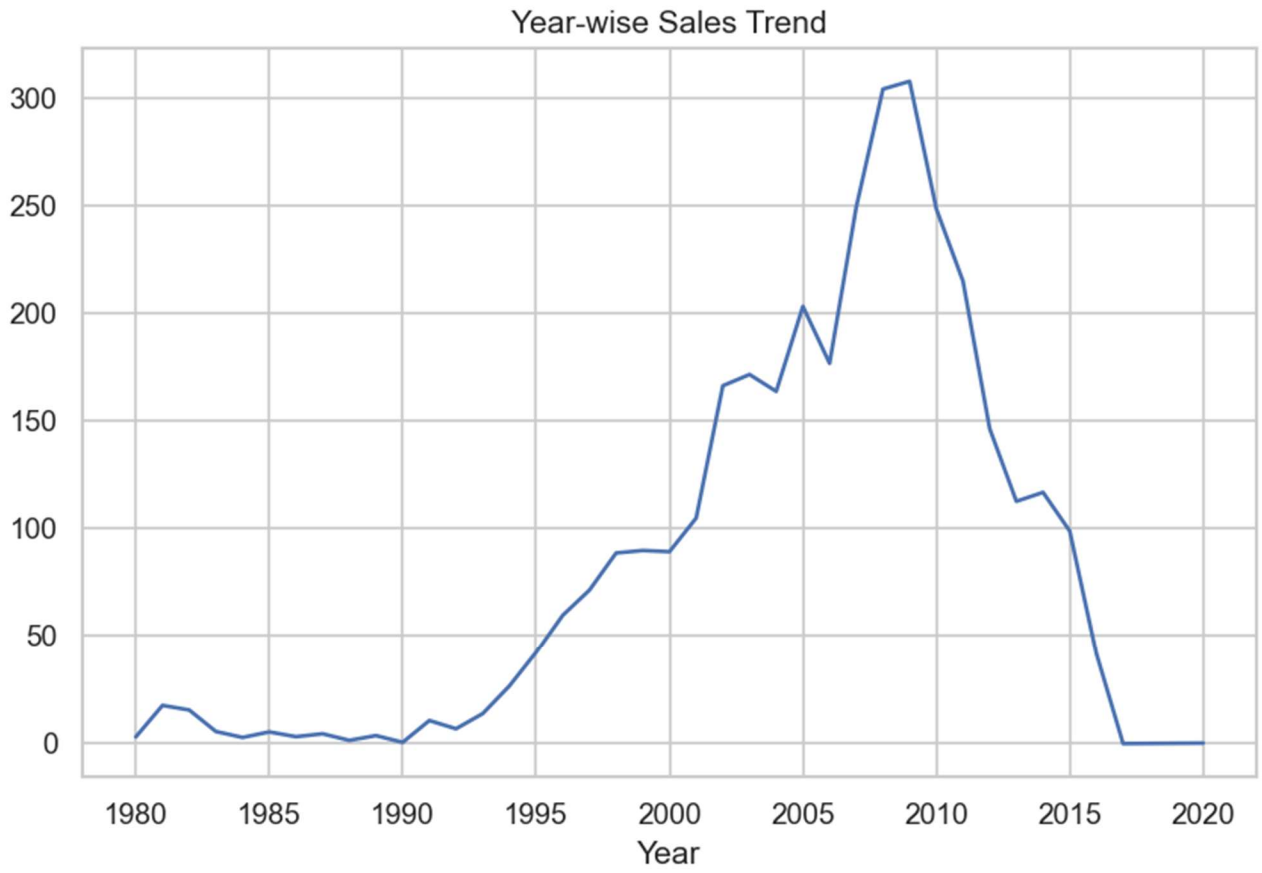

Density Curve of Sales
Density
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Global_Sales

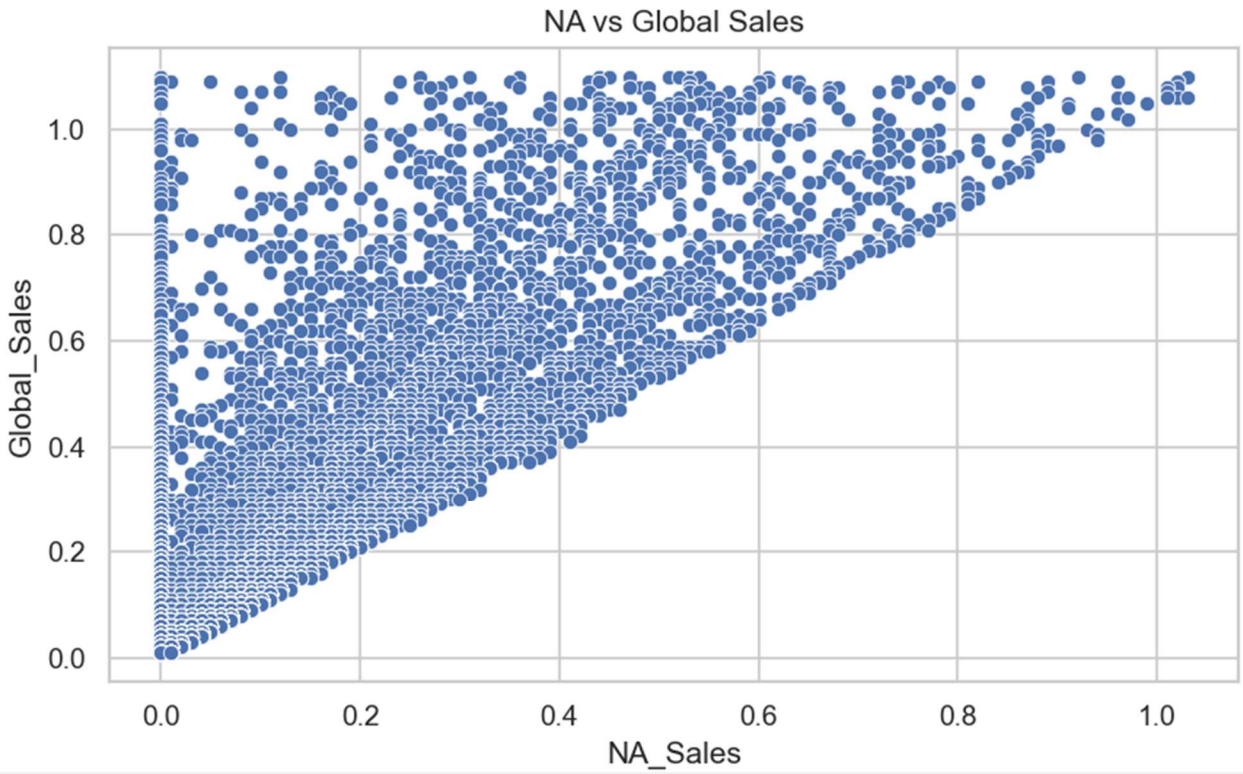

Top Genre Share
action
28.7%
sports
20.3%
misc
15.3%
role-playing
12.7%
adventure
12.3%
racing
10.6%

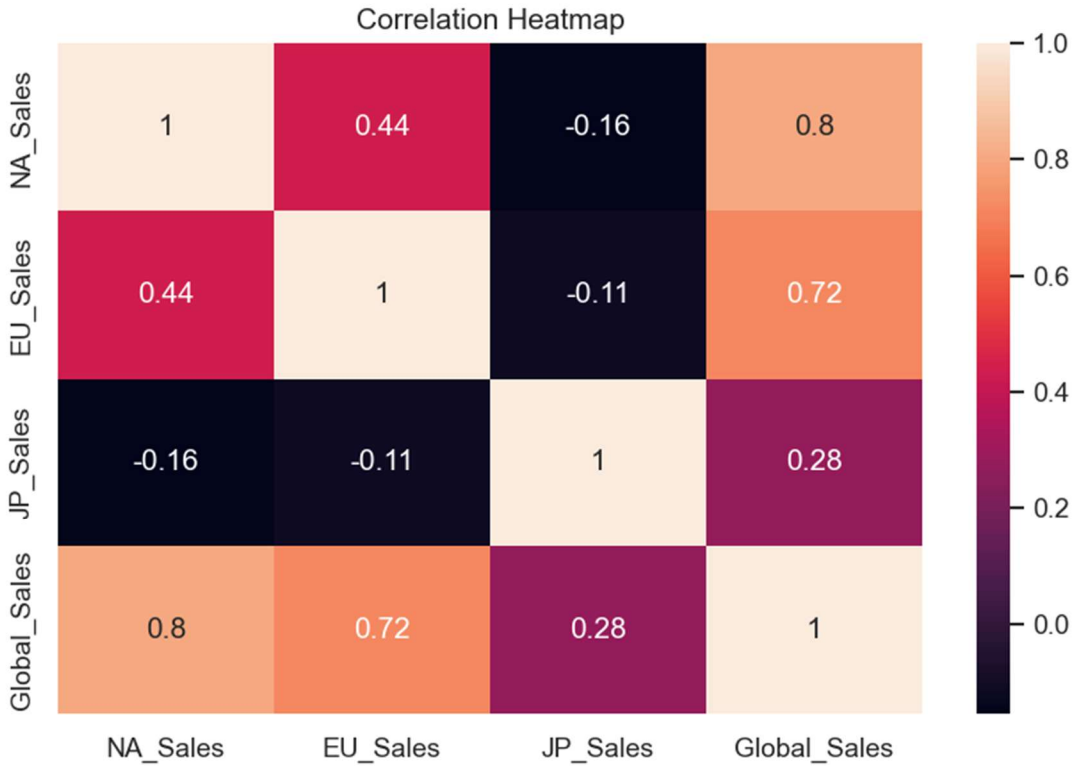

Top Platform Share
ds
24.3%
x360
12.4%
ps3
13.2%
psp
13.9%
wii
14.0%
ps2
22.2%


Top Publishers
400
350
300
250
200
150
100
50
0
c arts
vision
oisoft
thq
ment
ames
ment
endo
sega
ctive


Games by Genre
misc
shooter
action
platform
sports
simulation
fighting
strategy
puzzle
role-playing
racing
adventure
0
500
1000
1500
2000
2500
3000
count


Year-wise Sales Trend
300
250
200
150
100
50
0
1980
1985
1990
1995
2000
2005
2010
2015
2020
Year


NA vs Global Sales
Global_Sales
NA_Sales
0.0
0.2
0.4
0.6
0.8
1.0


Correlation Heatmap
NA_Sales
EU_Sales
JP_Sales
Global_Sales
1
0.44
-0.16
0.8
0.44
1
-0.11
0.72
-0.16
-0.11
1
0.28
0.8
0.72
0.28
1
NA_Sales
EU_Sales
JP_Sales
Global_Sales
1.0
0.8
0.6
0.4
0.2
0.0

```
Missing values after:
 Rank            0
Name             0
Platform         0
Year             0
Genre            0
Publisher        0
NA_Sales         0
EU_Sales         0
JP_Sales         0
Other_Sales      0
Global_Sales     0
dtype: int64

Final Shape after cleaning: (14465, 12)
===== DATA CLEANING COMPLETED =====


===== KPI SUMMARY =====
Total Games: 14465
Total Sales: 3389.45
Average Sales: 0.23432077428275144
Top Genre: action
Top Platform: ds

Correlation NA vs Global: 0.8017654134599297 P-value: 0.0

MODEL PERFORMANCE
MAE: 2.5396513587829844e-15
R2: 1.0

KEY INSIGHTS:
✔ Majority games have low sales
✔ Few games dominate revenue
✔ Action genre is most popular
✔ NA region leads in sales
✔ Strong regional correlation
```

```
    Rank                      Name Platform  ...  JP_Sales Other_Sales Global_Sa
les
0     1                Wii Sports      Wii  ...      3.77        8.46        82.
74
1     2          Super Mario Bros.      NES  ...      6.81        0.77        40.
24
2     3             Mario Kart Wii      Wii  ...      3.79        3.31        35.
82
3     4          Wii Sports Resort      Wii  ...      3.28        2.96        33.
00
4     5  Pokemon Red/Pokemon Blue       GB  ...     10.22        1.00        31.
37

[5 rows x 11 columns]

Last 5 Rows:
         Rank  ... Global_Sales
16593  16596  ...         0.01
16594  16597  ...         0.01
16595  16598  ...         0.01
16596  16599  ...         0.01
16597  16600  ...         0.01

[5 rows x 11 columns]

Columns:
 Index(['Rank', 'Name', 'Platform', 'Year', 'Genre', 'Publisher', 'NA_Sales',
       'EU_Sales', 'JP_Sales', 'Other_Sales', 'Global_Sales'],
      dtype='str')

Data Types:
 Rank              int64
Name                str
Platform            str
Year            float64
Genre               str
Publisher           str
NA_Sales        float64
EU_Sales        float64
JP_Sales        float64
```